

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of M. Sc. (AOC) Examination April 2018

OC 828: Medicinal Chemistry

Date: 30/04/2018 Day: Monday Time: 10.00 a.m. to 10.30 a.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q-I A Choose the correct answer for the following questions. 10

1 Phenothiazide is obtained by:

- (a) Cyclization of diphenylamine with sulfur
- (b) Cyclization of dibenzylamine with sulfur
- (c) Reduction of dibenzylamine with sulfur
- (d) All answer are wrong

2 Drug administered through the following route is most likely to be subjected to first-pass metabolism:

- (a) Oral (b) Sublingual (c) Subcutaneous (d) Rectal

3 Which of the following agents is related to tricyclic antidepressants?

- (a) Nefazodon (b) Amitriptyline (c) Fluoxetine (d) Isocarboxazid

- 4 Bioavailability of drug refers to:**
- (a) Percentage of administered dose that reaches systemic circulation in the unchanged form
 - (b) Ratio of oral to parenteral dose
 - (c) Ratio of orally administered drug to that excreted in the faeces
 - (d) Ratio of drug excreted unchanged in urine to that excreted as metabolites
- 5 Higher efficacy of a drug necessarily confers:**
- (a) Greater safety
 - (b) Therapeutic superiority
 - (c) Capacity to produce more intense response
 - (d) Cost saving
- 6 The essential feature in drug addiction is:**
- (a) Physical dependence
 - (b) Psychological dependence
 - (c) Both physical and psychological dependence
 - (d) Psychiatric abnormality
- 7 The combination of disulfiram and ethanol leads to accumulation of:**
- (a) Formaldehyde
 - (b) Acetate
 - (c) Formic acid
 - (d) Acetaldehyde
- 8 Compared to subcutaneous injection, the intramuscular injection of drugs:**
- (a) Is more painful
 - (b) Produces faster response
 - (c) Is unsuitable for depot preparations
 - (d) Carries greater risk of anaphylactic reaction
- 9 Which of the following groups of antibiotics demonstrates a bactericidal effect?**
- (a) Tetracyclines
 - (b) Macrolides
 - (c) Penicillins
 - (d) All of the above
- 10 Bacteriostatic effect is:**
- (a) Inhibition of bacterial cell division
 - (b) Inhibition of young bacterial cells growth
 - (c) Destroying of bacterial cells
 - (d) Formation of bacterial L-form

Q-I B Match A with B.

10

A		B		Answer
1	Vincristine Sulphate	A	Mustards	
2	Isosorbide Dinitrate	B	Anti-Arrhythmic drug	
3	Actinomycin-D	C	4-[4-bis(2-chloroethyl)amino phenyl] butyric acid	
4	Guanethidine	D	Antihyper cholesterolemic drug	
5	Prazocin	E	Vasodilators	
6	Sulphamethoxazole	F	ACE inhibitor	
7	5-fluoro Uracil	G	Cytosan	
8	Diazoxide	H	Antineoplastic Antibiotic	
9	Nifedipine	I	Loop diuretics	
10	Azathiopurine	J	Thiazide diuretics	
11	Methotrexate	K	Used for Acute leukemia in Hodgkin's disease	
12	Hydrochlorothiazide	L	1-(2-chloroethyl)-3-cyclo hexyl-1-nitroso urea	
13	Lomustine	M	Adrenergic neuron blockers	
14	Furosemide	N	Inhibitor of DHFR	
15	Cyclophosphamide	O	Pyrimidine Antagonist	
16	Captopril	P	Direct vasodilation	
17	Lovastatin	Q	α – Adrenergic blockers	
18	Chlorambucil	R	Calcium Antagonist	
19	Verapamil	S	Purine Antagonist	
20	Melphalan	T	Act as antimetabolites	

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OC 828: Medicinal Chemistry

Date: 30/04/2018 Day: Monday Time: 10.31 a.m. to 01.00 p.m. Maximum Marks: 50

Important Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
 - Make suitable assumptions and draw neat figures wherever required.
 - Use of non-programmable calculator is allowed.
 - Show necessary calculations.
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SECTION-I

Q-II Answer the following. 10

- A** Justify: in case of non-catalytic protein receptor, a drug salbutamol act as an agonist drug and a drug ranitidine act as an antagonist drug. 4
- B** Write a brief note on: Angiotension converting enzyme inhibitor. 2

OR

- B** Give the difference between IV and IM administration of drug. 2
- C** List the factors affecting drug absorption and explain them in detail. 4

OR

- C** Give a synthesis and clinicalness of Chloramphenicol. 4

Q-III Answer the following. 10

- A** Give an account of β -lactumase inhibitors. 4
- B** Discuss the chemical reaction of penicillin. 3
- C** Explain: LDL is bad cholesterol while HDL is good cholesterol. 3

OR

- C** Write a note on: Calcium channel blocker with suitable example. 3

SECTION-II

Q-IV Give the synthesis of the following drugs. (Any Five) 10

- (i) Guanethidine (ii) Prazocin (iii) Diazoxide (iv) Chlorothiazide
(v) Melphalan (vi) 5-fluoro uracil (vii) Amitriptyline (viii) Phenytein

Q-V Answer the following. 10

- A** Justify: Sulphamethoxazole act as antimetabolite. 4
B Explain: The methotrexate is used as a folic acid antagonist. 4
C Write the structure and clinicalness of following molecules. **(Any Two)** 2
(i) Isosorbide Dinitrate (ii) Cytarabine (iii) Desipramine

Q-VI Answer the following. 10

- A** Write a note on: Prodrug 3
B Write a note on: Biotransformation 3
C Discuss the structure –functions relationship among Phenothiazide with respects to, Modification of alkyl side chains at position-10. 4

OR

- C** Give an account of dibenzocycloheptadiene. 4